Covariance Matrix

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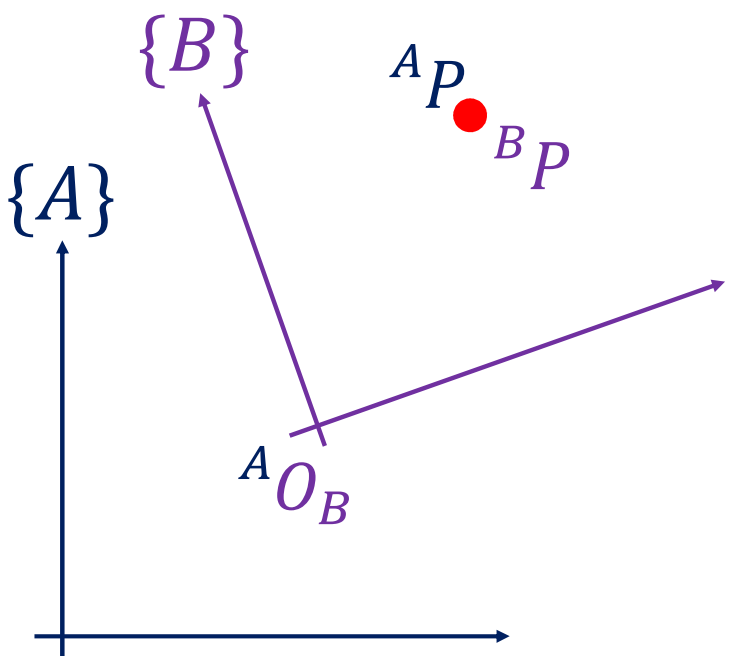
Overview

- Coordinate Transformations
- Recalling Definitions
  - Covariance Matrix
  - Mahalanobis Distance
  - Matrix Diagonalization
- Covariance Matrix – An Interpretation



Coordinate Transformations

Coordinate Transformations: Notations

Lagging Superscript

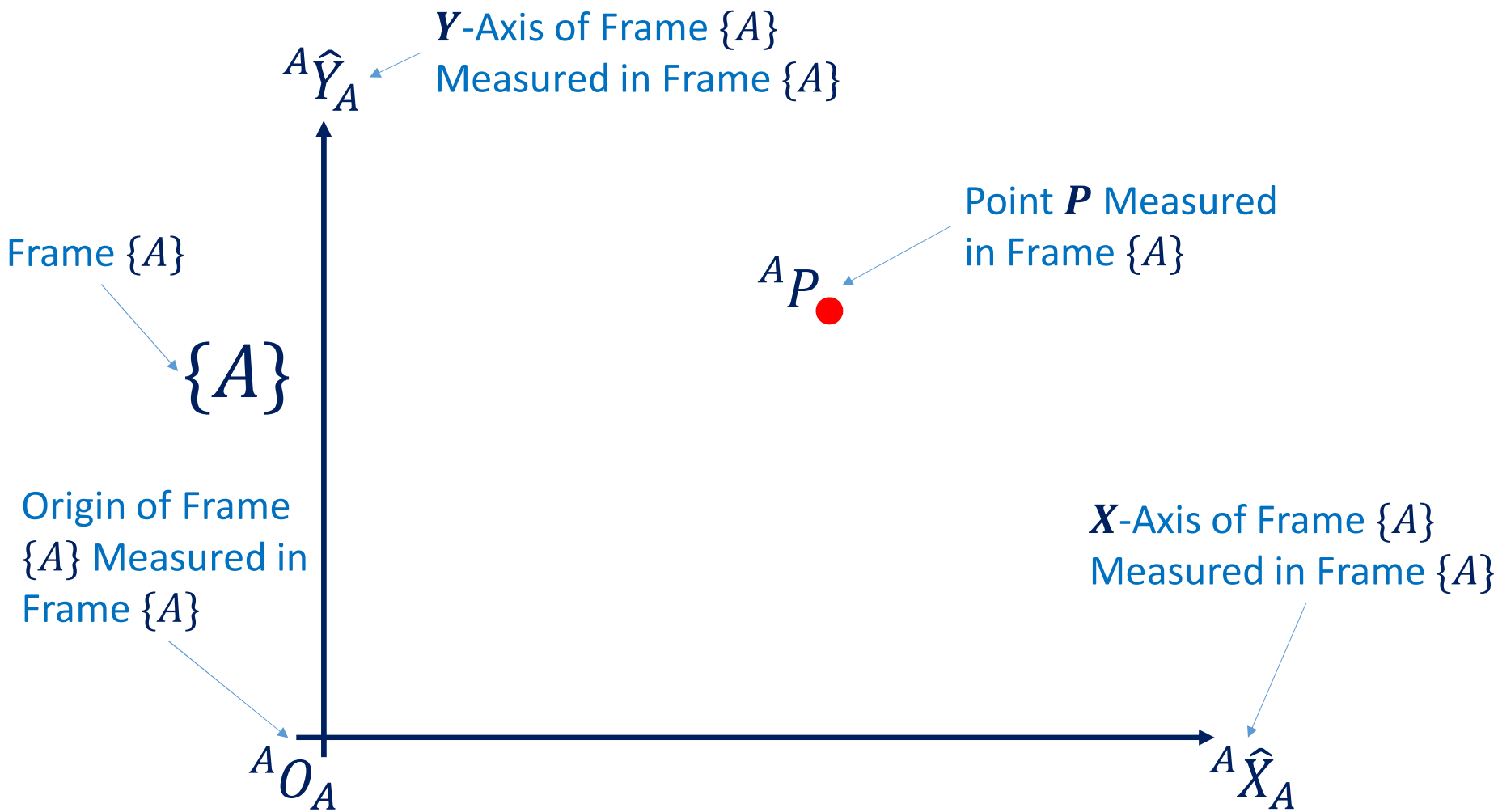
Leading Superscript

$$ {}^{A}_{B}P^{C}_{D} $$

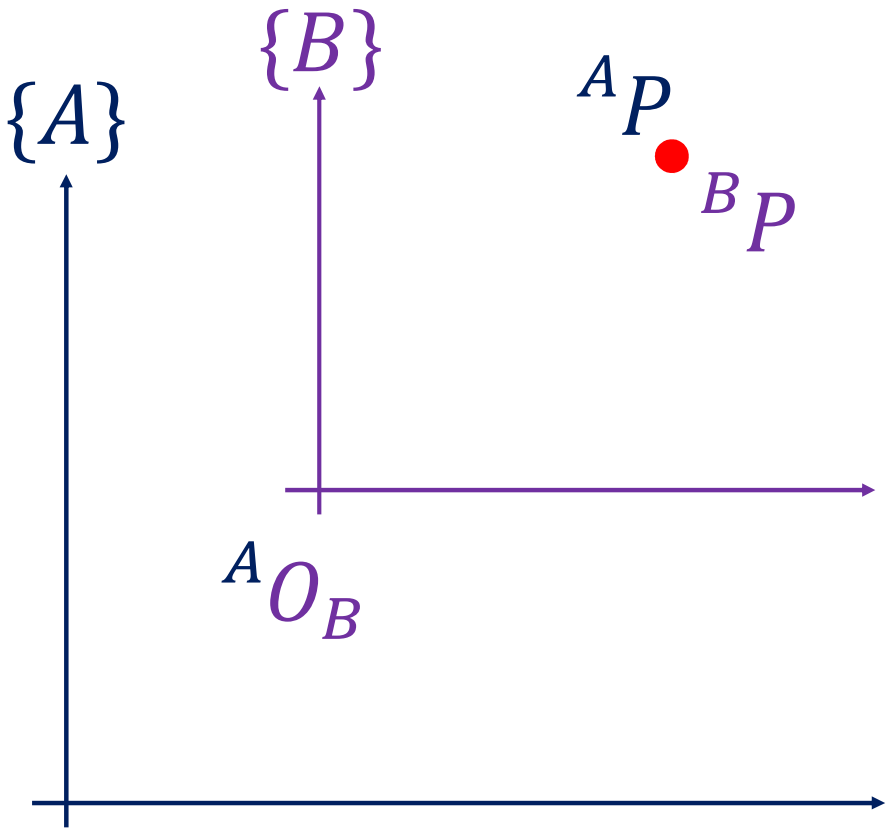
Lagging Subscript

Leading Subscript

Coordinate Transformations: Notations



Coordinate Transformations: Translation

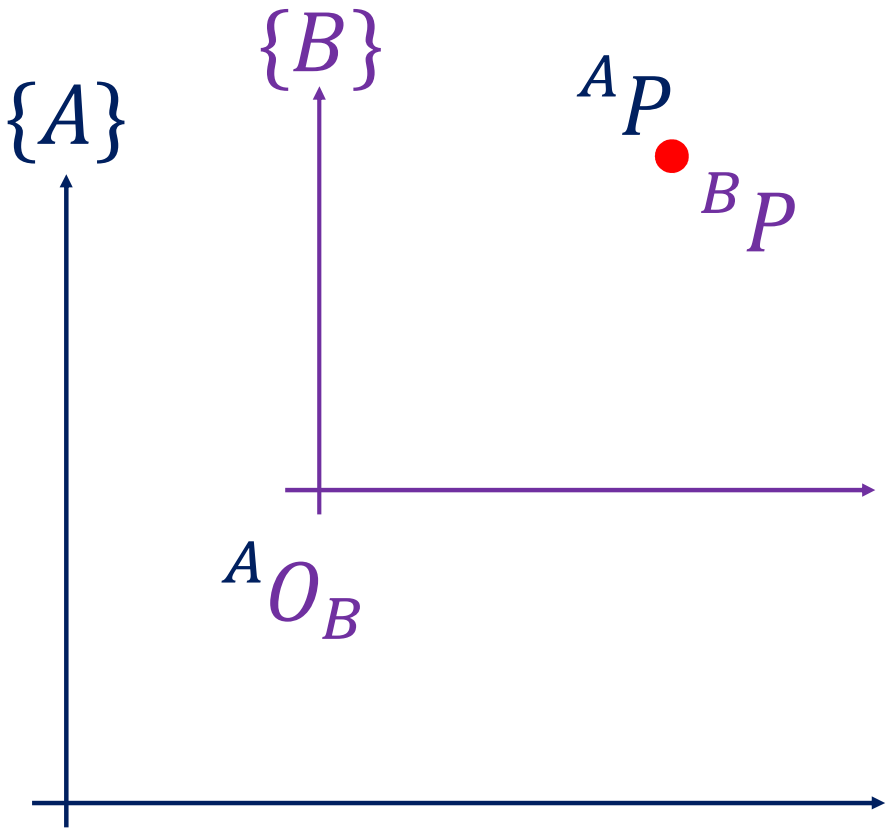


$${}^{A}\boldsymbol{P} = (10,10)$$

$${}^{A}O_B = (3,5)$$

$${}^{B}\boldsymbol{P} = ?$$

Coordinate Transformations: Translation



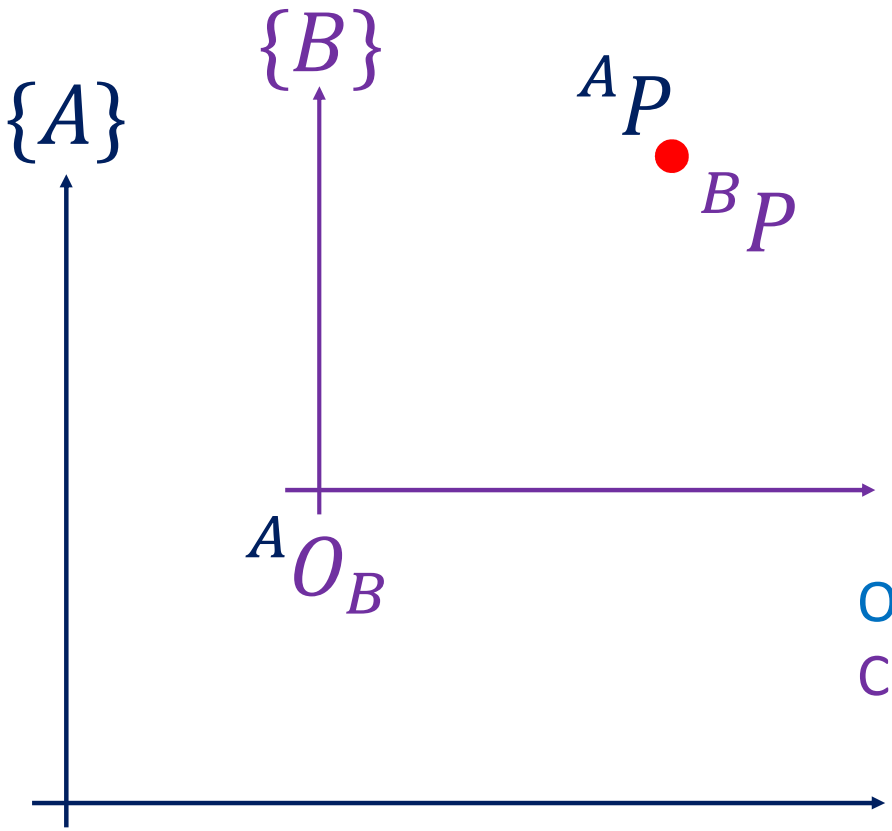
$${}^{A}\boldsymbol{P} = (10,10)$$

$${}^{A}O_B = (3,5)$$

$${}^{B}\boldsymbol{P} = (10,10) - (3,5)$$

$${}^{B}\boldsymbol{P} = (7,5)$$

Coordinate Transformations: Translation



$$ {}^{B}\boldsymbol{P} = {}^{A}\boldsymbol{P} - {}^{A}\boldsymbol{O}_{\boldsymbol{B}} $$

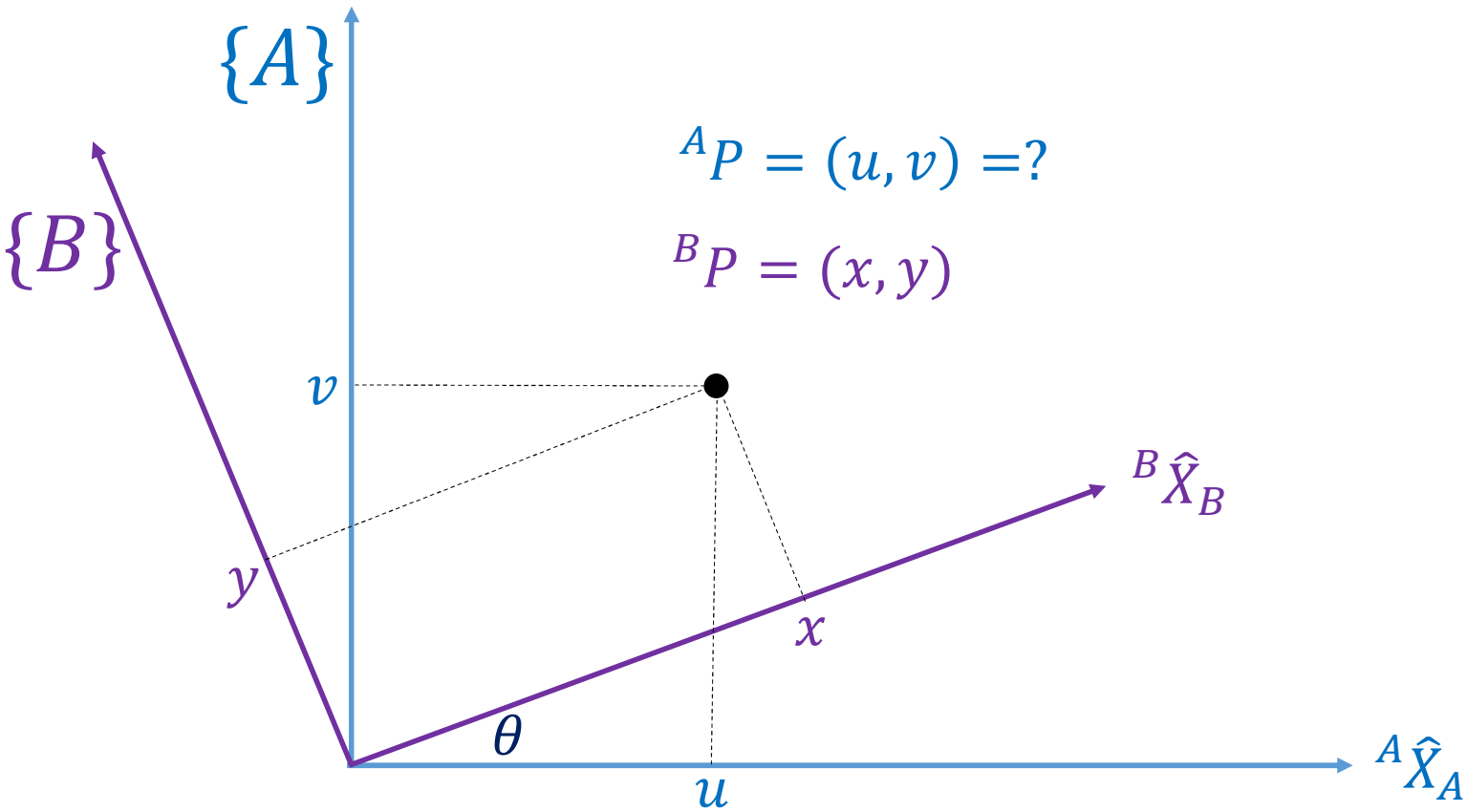
Translation

Origins of *{A}* and *{B}* do not coincide

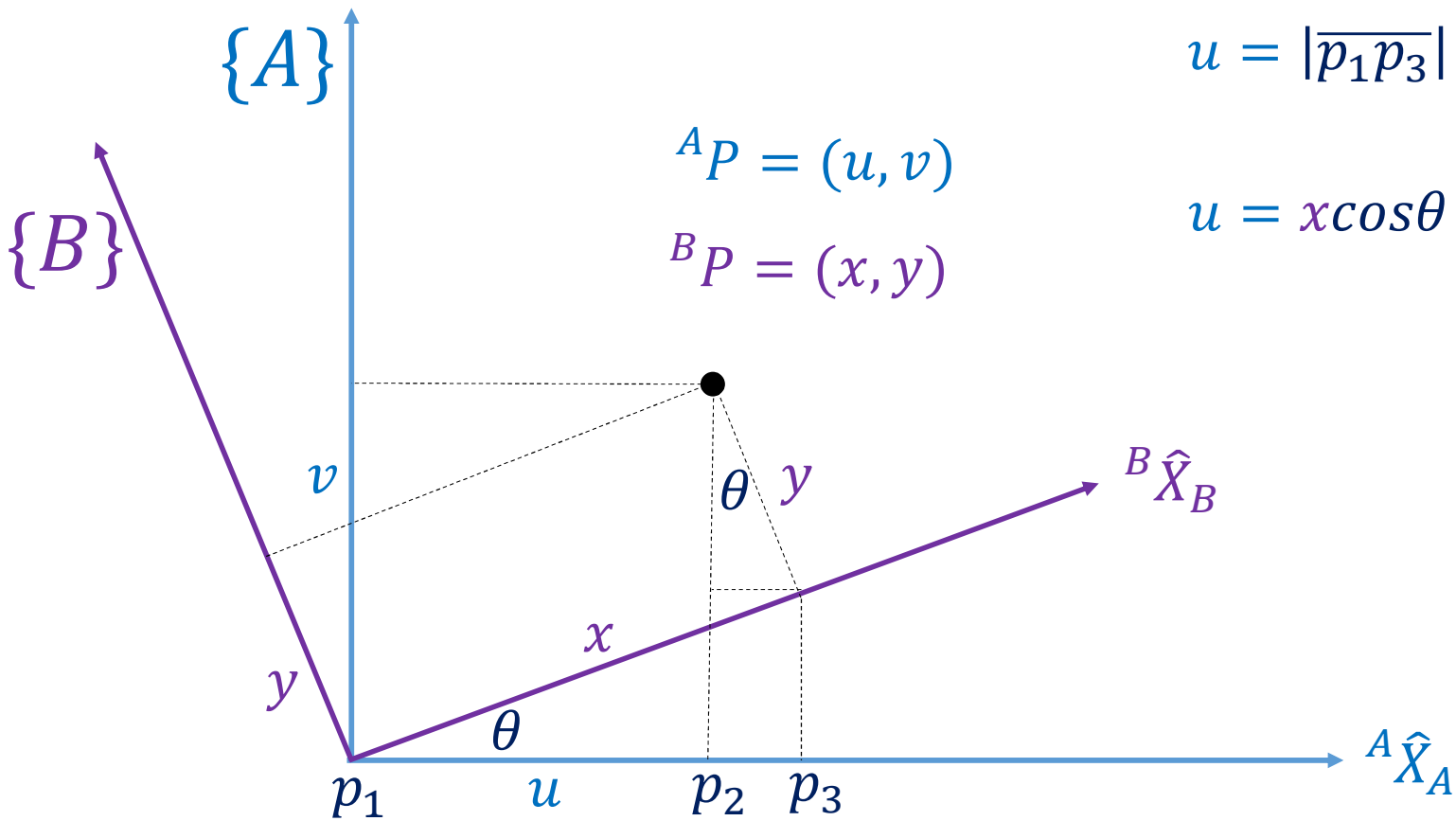
Corresponding Axes of *{A}* and *{B}* are parallel

Everything on R.H.S. are Measured w.r.t. *{A}*

Coordinate Transformations: Rotation



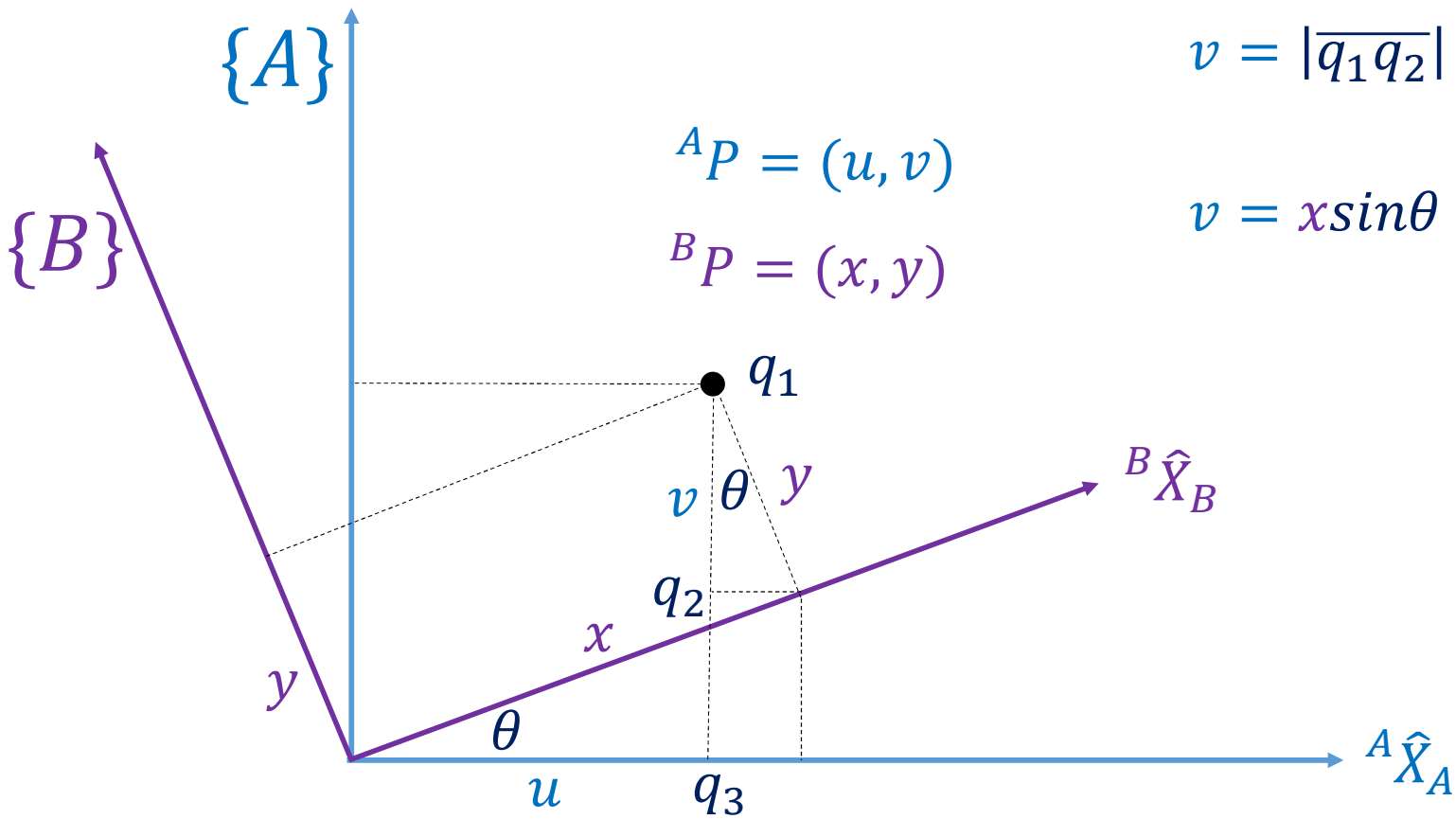
Coordinate Transformations: Rotation



$$u = |\overline{p_1 p_3}| - |\overline{p_2 p_3}|$$

$$u = x\cos\theta - y\sin\theta$$

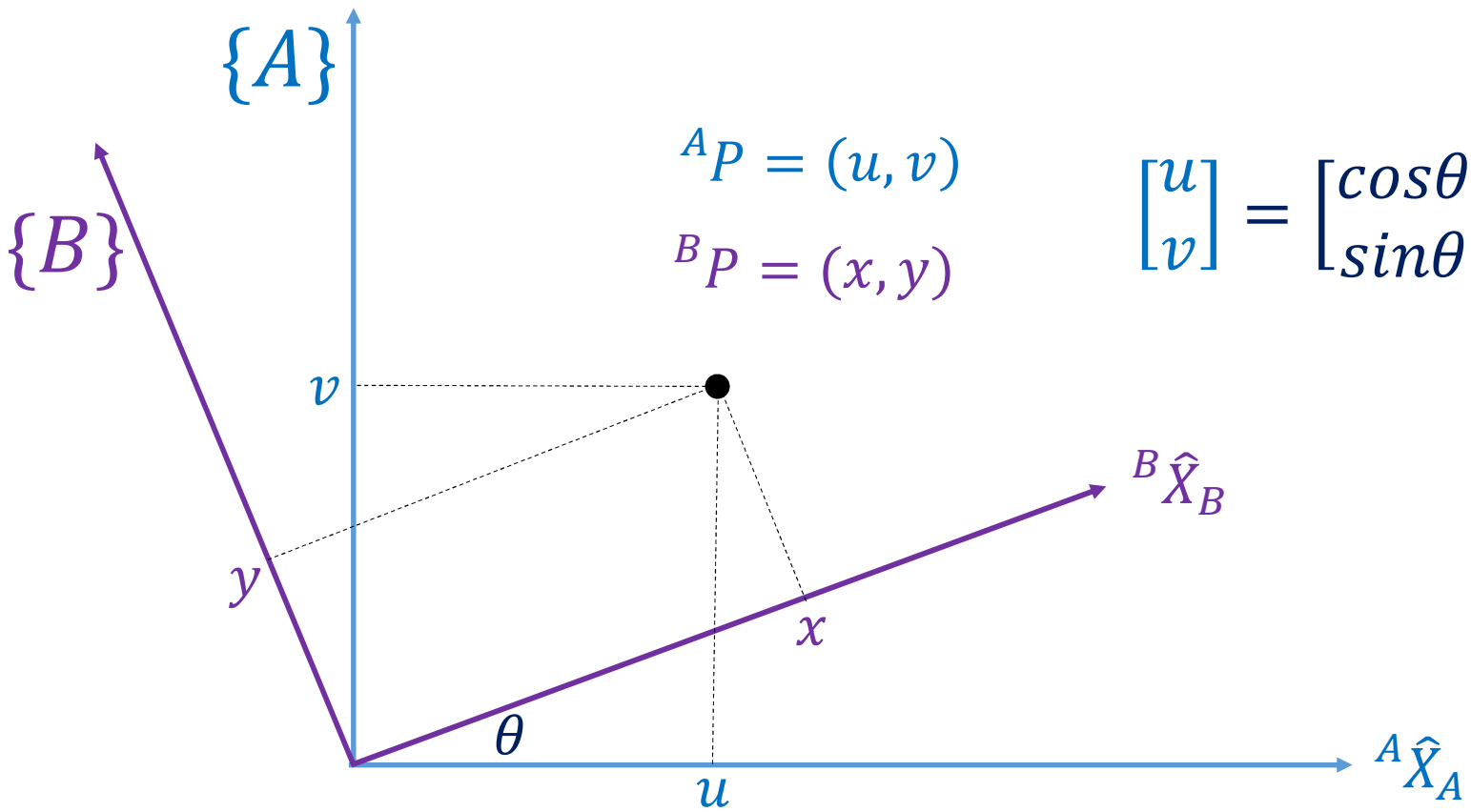
Coordinate Transformations: Rotation



$$v = |\overline{q_1 q_2}| + |\overline{q_2 q_3}|$$

$$v = x sin\theta + y cos\theta$$

Coordinate Transformations: Rotation



$${}^{A}P = (u, v)$$

$${}^{B}P = (x, y)$$

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Rotation Matrix in 2D

$$R = \begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix}$$

$$|R| = ?$$

$$R^{-1} = ?$$

Rotation Matrix in 2D

$$R = \begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix}$$

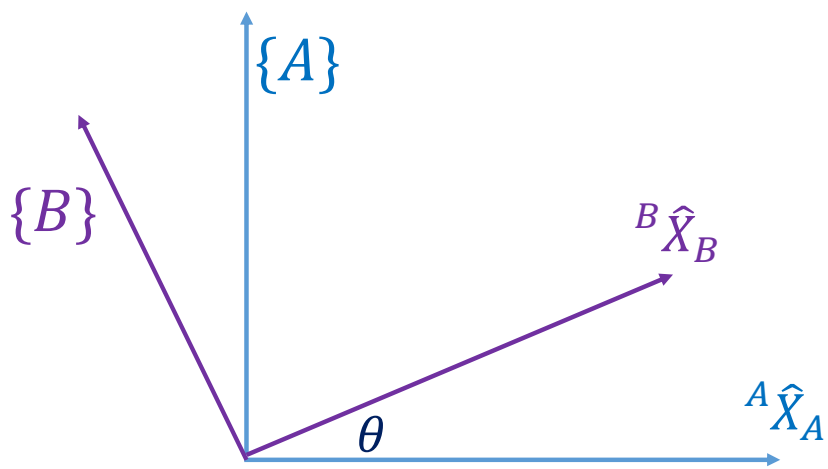
$$|R| = 1$$

$$R^{-1} = \begin{bmatrix} cos\theta & sin\theta \\ -sin\theta & cos\theta \end{bmatrix} = R^T$$

Coordinate Transformations: Rotation

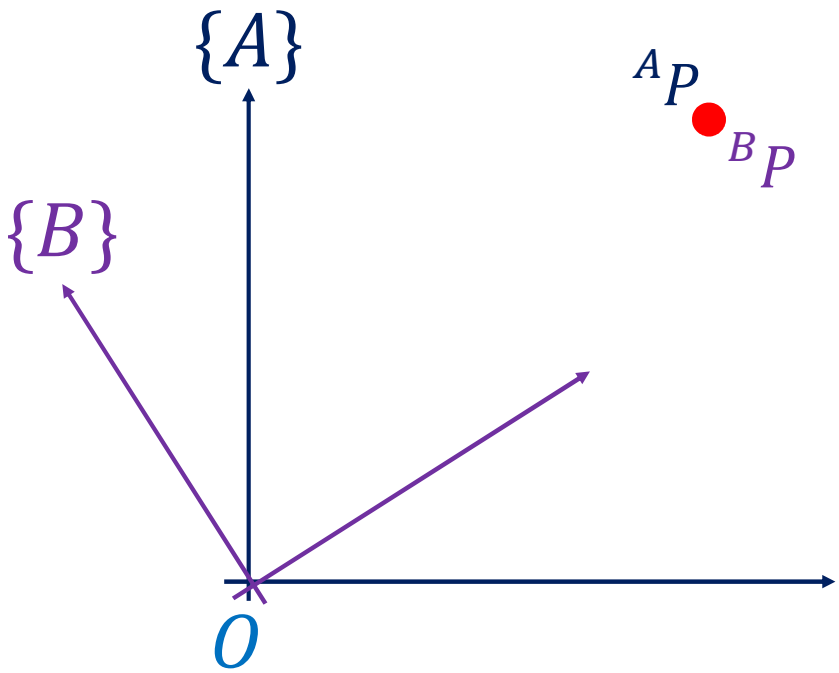
$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \longrightarrow {}^{A}P = {}^{A}_{B}R\ {}^{B}P$$

$$\left| {}^{A}_{B}R \right| = 1 \qquad \left\langle \begin{bmatrix} cos\theta \\ sin\theta \end{bmatrix}, \begin{bmatrix} -sin\theta \\ cos\theta \end{bmatrix} \right\rangle = 0 \qquad {}^{A}_{B}R^{\mathrm{T}} = {}^{A}_{B}R^{-1}$$



$${}^{A}\hat{X}_{B} = \begin{bmatrix} cos\theta \\ sin\theta \end{bmatrix} \qquad {}^{A}\hat{Y}_{B} = \begin{bmatrix} -sin\theta \\ cos\theta \end{bmatrix}$$

Coordinate Transformations: Rotation



$$^{A}P = {}^{A}_{B}R\ {}^{B}P$$

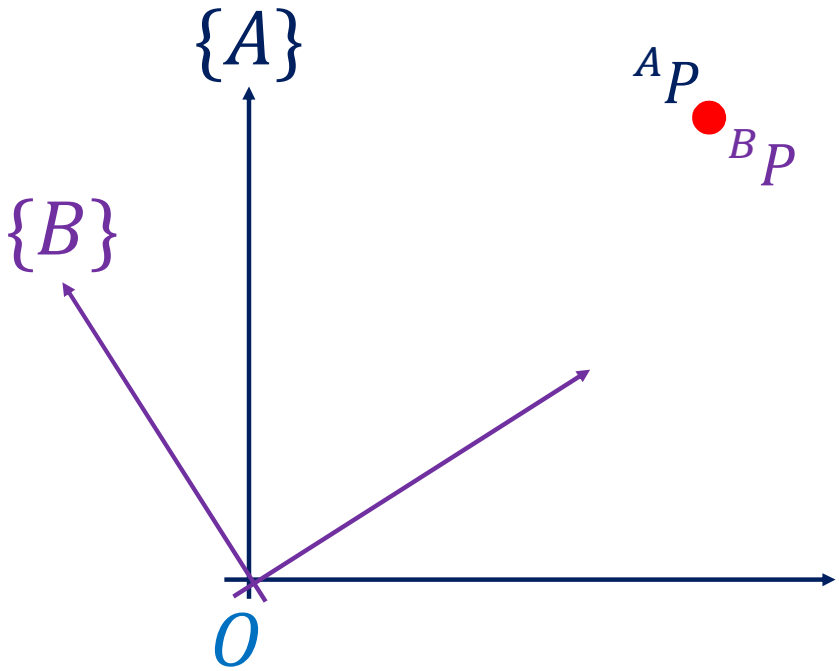
Application of Rotation Matrix ${}^{A}_{B}R$
From Reference Frame $\{B\}$ to Frame $\{A\}$

$$^{A}\hat{X}_B = {}^{A}_{B}R\ {}^{B}\hat{X}_B \Rightarrow {}^{A}\hat{X}_B = {}^{A}_{B}R \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$^{A}\hat{Y}_B = {}^{A}_{B}R\ {}^{B}\hat{Y}_B \Rightarrow {}^{A}\hat{Y}_B = {}^{A}_{B}R \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} {}^{A}\hat{X}_B & {}^{A}\hat{Y}_B \end{bmatrix} = {}^{A}_{B}R \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Coordinate Transformations: Rotation



$$^{A}P = {}^{A}_{B}R\ {}^{B}P$$

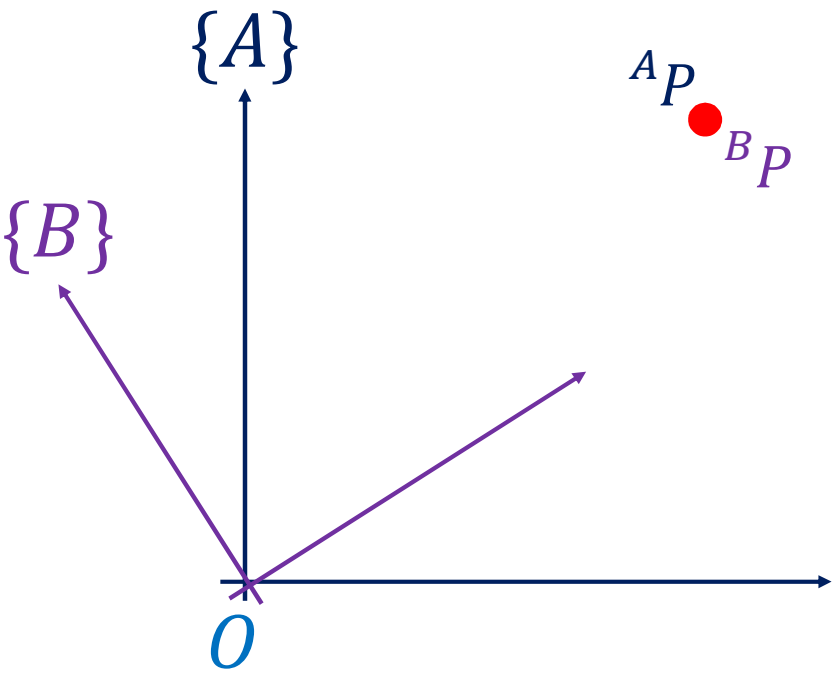
Application of Rotation Matrix ${}^{A}_{B}R$
From Reference Frame $\{B\}$ to Frame $\{A\}$

$$\begin{bmatrix} {}^{A}\hat{X}_{B} & {}^{A}\hat{Y}_{B} \end{bmatrix} = {}^{A}_{B}R$$

Axes of Frame $\{B\}$ (Source) Measured in Frame $\{A\}$ (Destination)

Transformation to Frame $\{A\}$ (Destination) from Frame $\{B\}$ (Source)

Coordinate Transformations: Rotation



$${}^{A}_{B}R = \left[\, {}^{A}\hat{X}_B \quad {}^{A}\hat{Y}_B \,\right]$$

$${}^{B}_{A}R = \left[\, {}^{B}\hat{X}_A \quad {}^{B}\hat{Y}_A \,\right]$$

$${}^{A}P = {}^{A}_{B}R\ {}^{B}P$$

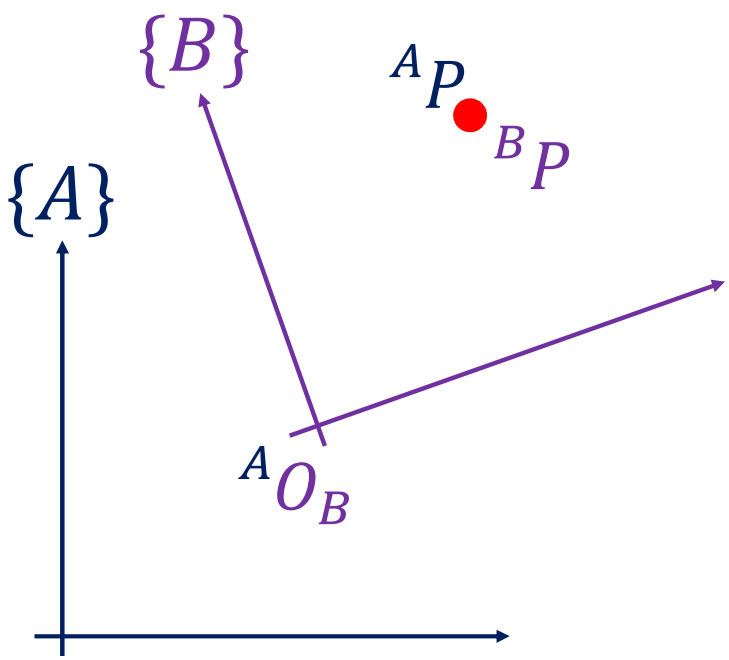
Application of Rotation Matrix ${}^{A}_{B}R$
From Reference Frame $\{B\}$ to Frame $\{A\}$

$${}^{B}P = {}^{A}_{B}R^{-1}\ {}^{A}P$$

$${}^{A}_{B}R^{-1}\ {}^{A}P = {}^{A}_{B}R^{T}\ {}^{A}P$$

$${}^{A}_{B}R^{T}\ {}^{A}P = {}^{B}_{A}R\ {}^{A}P$$

Coordinate Transformations: Translation & Rotation



Given: ${}^{A}P$

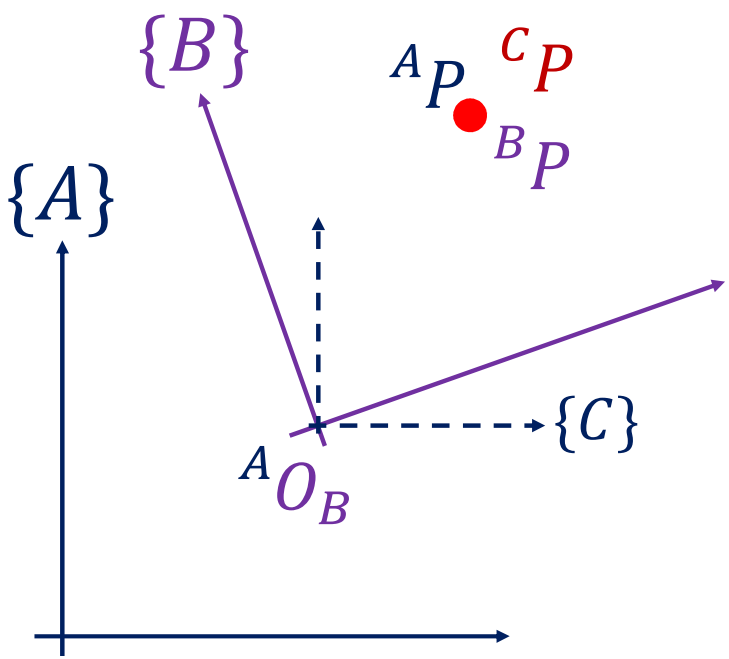
Evaluate: ${}^{B}P$

With Knowledge of: $\{A\}, \{B\}$

- Only Translation
  - Origins of $\{A\}$ and $\{B\}$ Do Not Coincide
  - Corresponding Axes of $\{A\}$ and $\{B\}$ are Parallel

- Only Rotation
  - Origins of $\{A\}$ and $\{B\}$ Must Coincide
  - Corresponding Axes of $\{A\}$ and $\{B\}$ are Not Parallel

Coordinate Transformations: Translation & Rotation



Construct a Temporary Frame $\{C\}$

- ❑ Origins of $\{A\}$ and $\{C\}$ Do Not Coincide
- ❑ Corresponding Axes of $\{A\}$ and $\{C\}$ are Parallel
- ❑ Origins of $\{C\}$ and $\{B\}$ Coincide
- ❑ Corresponding Axes of $\{C\}$ and $\{B\}$ are Not Parallel

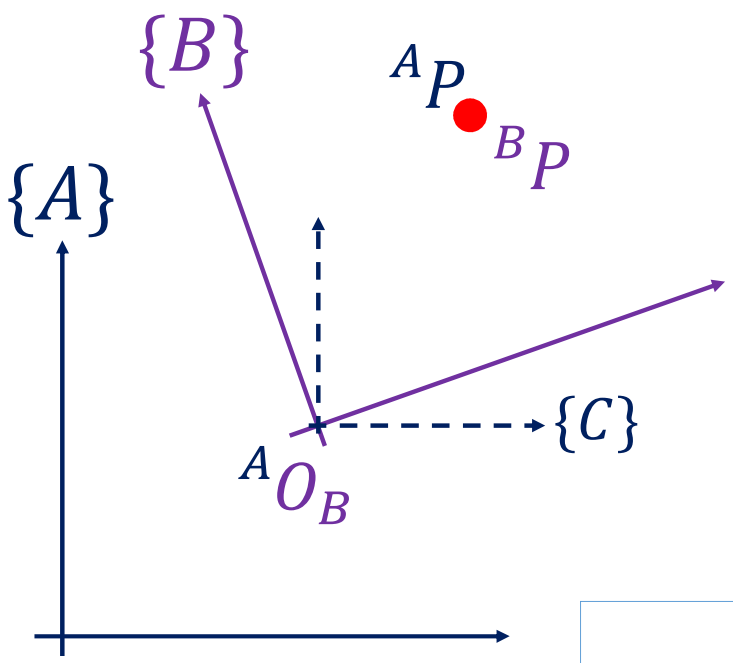
Translate First

$$ {}^{C}P = {}^{A}P - {}^{A}O_C $$

Observation

$$ {}^{A}O_C = {}^{A}O_B $$

Coordinate Transformations: Translation & Rotation



Construct a Temporary Frame $\{C\}$

- ❑ Origins of $\{A\}$ and $\{C\}$ Do Not Coincide
- ❑ Corresponding Axes of $\{A\}$ and $\{C\}$ are Parallel
- ❑ Origins of $\{C\}$ and $\{B\}$ Coincide
- ❑ Corresponding Axes of $\{C\}$ and $\{B\}$ are Not Parallel

Rotate Next

$${}^{B}P = {}^{B}_{C}R\ {}^{C}P$$

Observation

$${}^{B}_{C}R = {}^{B}_{A}R = {}^{A}_{B}R^{-1} = {}^{A}_{B}R^{\mathrm{T}}$$

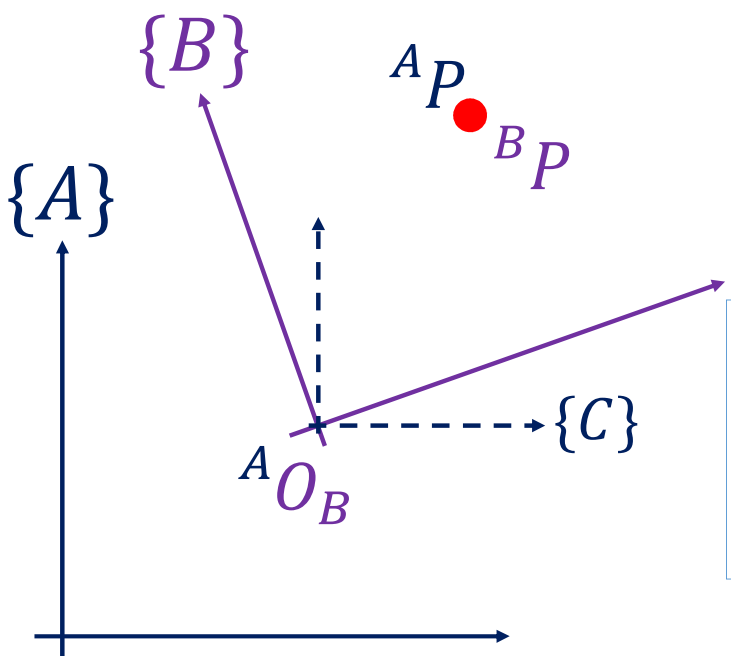
Direction Measurement w.r.t. $\{A\}$ and $\{C\}$ are equivalent

All Columns are Measured w.r.t. $\{A\}$

$${}^{B}_{C}R = \left[\ {}^{B}\hat{X}_C \quad {}^{B}\hat{Y}_C \quad \cdots\right] = \left[\ {}^{B}\hat{X}_A \quad {}^{B}\hat{Y}_A \quad \cdots\right] = {}^{B}_{A}R$$

$${}^{A}_{B}R = \left[\ {}^{A}\hat{X}_B \quad {}^{A}\hat{Y}_B \quad \cdots\right]$$

Coordinate Transformations: Translation & Rotation



$${}^{C}P = {}^{A}P - {}^{A}O_C \qquad {}^{A}O_C = {}^{A}O_B$$

$${}^{C}P = {}^{A}P - {}^{A}O_B$$

$${}^{B}P = {}^{B}_{C}R\ {}^{C}P \qquad {}^{B}_{C}R = {}^{B}_{A}R$$

$${}^{B}P = {}^{B}_{A}R\ {}^{C}P$$

$${}^{B}P = {}^{B}_{A}R\left({}^{A}P - {}^{A}O_B\right)$$

$${}^{B}P = {}^{A}_{B}R^{T}\left({}^{A}P - {}^{A}O_B\right)$$

Quick Revision

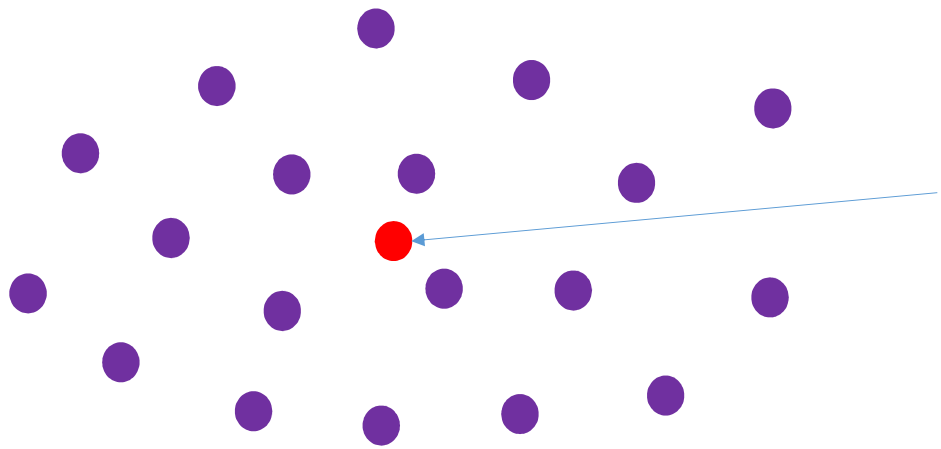
- Covariance Matrix
- Mahalanobis Distance
- Matrix Diagonalization

Scalar Dispersion d_{mv} for Multivariate Data

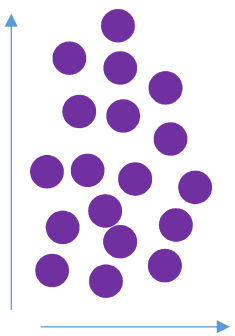
$\boldsymbol{x}_i \in \mathbb{R}^d$

$\boldsymbol{\mu} \in \mathbb{R}^d$

$d_{mv} \in \mathbb{R}^1$



$$\boldsymbol{\mu} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i$$



Different Dispersions
in Different Directions

$$d_{mv}^2 = \frac{1}{n}\sum_{i=1}^{n} \|x_i - \mu_A\|_2^2$$

d_{mv} Does Not Capture Directionality of Dispersion

Covariance $\boldsymbol{Cov(X,Y)}$

$$X = \{x_1, \dots, x_i, \dots, x_n\}$$

$$\Downarrow$$

$$\mu_X$$

$$Y = \{y_1, \dots, y_i, \dots, y_n\}$$

$$\Downarrow$$

$$\mu_Y$$

$$Cov(X,Y) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu_X)(y_i - \mu_Y) = \left(\frac{1}{n}\sum_{i=1}^{n} x_i y_i\right) - \mu_x \mu_y$$

$$Cov(X,Y) = Cov(Y,X)$$

Covariance Matrix ($\boldsymbol{C}$)

$$\boldsymbol{X} = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_i, \dots, \boldsymbol{x}_n\} \longrightarrow \boldsymbol{x} \in \mathbb{R}^d$$

$$\boldsymbol{X}_r = \{\boldsymbol{x}_1[r], \dots, \boldsymbol{x}_i[r], \dots, \boldsymbol{x}_n[r]\} \longrightarrow r = 1, \dots d$$

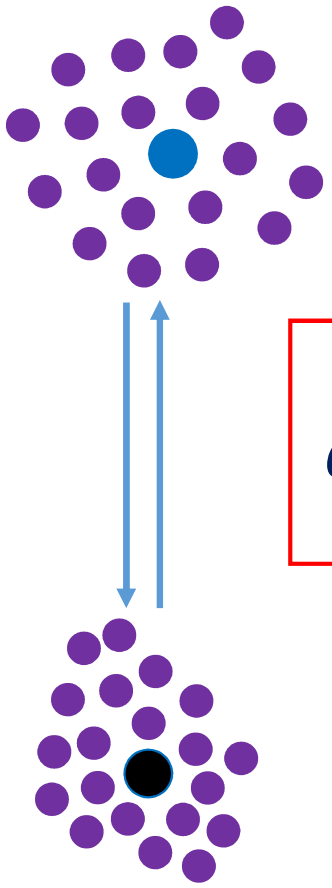
$$\boxed{\boldsymbol{C} = \frac{1}{n}\sum_{i=1}^{n}(\boldsymbol{x}_i - \boldsymbol{\mu})(\boldsymbol{x}_i - \boldsymbol{\mu})^T}$$

$$C = C^T$$

$$C[j][k] = C[k][j] = Cov(\boldsymbol{X}_j, \boldsymbol{X}_k) = Cov(\boldsymbol{X}_k, \boldsymbol{X}_j)$$

Mahalanobis Distance

$\boldsymbol{S_2} \equiv (\boldsymbol{\mu_2}, \boldsymbol{C_2})$



$\boldsymbol{S_1} \equiv (\boldsymbol{\mu_1}, \boldsymbol{C_1})$

$$d^2(\boldsymbol{\mu_2}, \boldsymbol{S_1}) = (\boldsymbol{\mu_2} - \boldsymbol{\mu_1})^T \boldsymbol{C_1^{-1}} (\boldsymbol{\mu_2} - \boldsymbol{\mu_1})$$

$$\boxed{d^2(\boldsymbol{S_1}, \boldsymbol{S_2}) = \frac{1}{2} (\boldsymbol{\mu_1} - \boldsymbol{\mu_2})^T \left(\boldsymbol{C_1^{-1}} + \boldsymbol{C_2^{-1}}\right) (\boldsymbol{\mu_1} - \boldsymbol{\mu_2})}$$

$$d^2(\boldsymbol{\mu_1}, \boldsymbol{S_2}) = (\boldsymbol{\mu_1} - \boldsymbol{\mu_2})^T \boldsymbol{C_2^{-1}} (\boldsymbol{\mu_1} - \boldsymbol{\mu_2})$$

Mahalanobis Distance

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

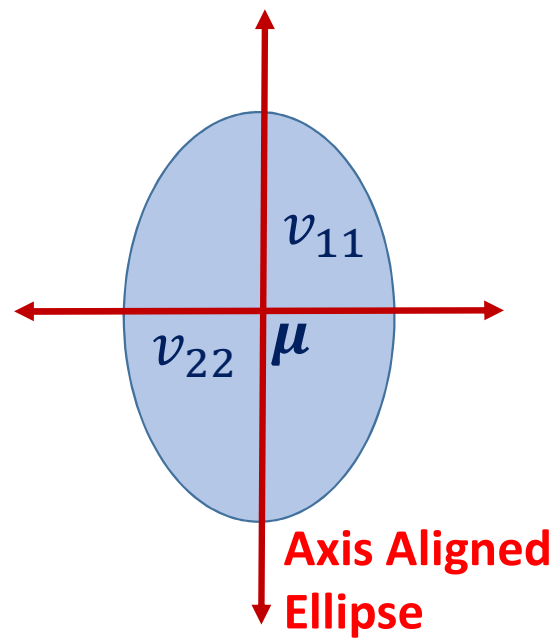
$$\boldsymbol{S} \equiv (\boldsymbol{\mu}, \boldsymbol{C})$$

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

$$\boldsymbol{C} = \begin{bmatrix} v_{11} & 0 \\ 0 & v_{22} \end{bmatrix}$$

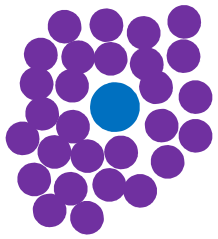
$$d^2(\boldsymbol{x}, \boldsymbol{S}) = (\boldsymbol{x} - \boldsymbol{\mu})^T \boldsymbol{C}^{-1} (\boldsymbol{x} - \boldsymbol{\mu})$$

$$\boxed{d^2(\boldsymbol{x}, \boldsymbol{S}) = \frac{(x_1 - \mu_1)^2}{v_{11}} + \frac{(x_2 - \mu_2)^2}{v_{22}}}$$



Mahalanobis Distance

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

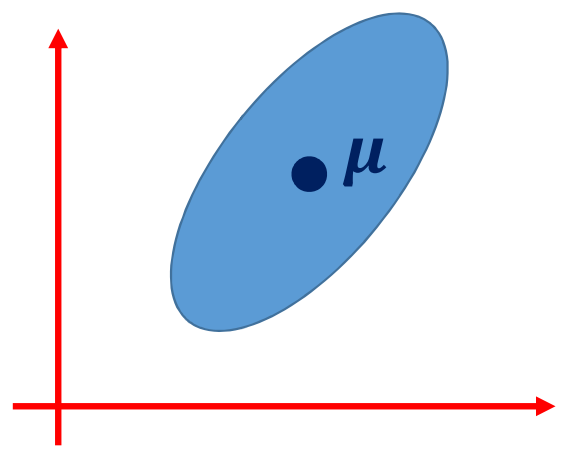


$$\boldsymbol{S} \equiv (\boldsymbol{\mu}, \boldsymbol{C})$$

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

$$\boldsymbol{C} = \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix}$$

$$\boxed{d^2(\boldsymbol{x}, \boldsymbol{S}) = a(x_1 - \mu_1)^2 + 2b(x_1 - \mu_1)(x_2 - \mu_2) + c(x_2 - \mu_2)^2}$$



Eigen Values and Eigen Vectors of Matrix C

Matrix C of size $d \times d$

$$C = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1d} \\ a_{21} & a_{22} & \cdots & a_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ a_{d1} & a_{d2} & \cdots & a_{dd} \end{bmatrix}$$

Eigen Vectors of C

$$\hat{\boldsymbol{e}}_1, \ldots \hat{\boldsymbol{e}}_i, \ldots \hat{\boldsymbol{e}}_d$$

$$\lambda_1, \ldots \lambda_i, \ldots \lambda_d$$

Eigen Values of C

Diagonal Matrix of Eigen Values

$$\hat{C} = \begin{bmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \lambda_d \end{bmatrix}$$

Matrix of Eigen Vectors

$$Q = [\hat{\boldsymbol{e}}_1 \quad \cdots \quad \hat{\boldsymbol{e}}_i \quad \cdots \quad \hat{\boldsymbol{e}}_d]$$

Matrix Diagonalization

$$C = Q\hat{C}Q^{-1}$$

$$C = Q\hat{C}Q^{T}$$

$$C = C^T$$

$$Q^{-1} = Q^T$$

$$|Q| = 1$$

$$\hat{\boldsymbol{e}}_i{}^T\hat{\boldsymbol{e}}_j = \delta[i-j]$$

$$C = \sum_{i=1}^{d} \lambda_i \hat{\boldsymbol{e}}_i \hat{\boldsymbol{e}}_i^T = \lambda_1 \hat{\boldsymbol{e}}_1 \hat{\boldsymbol{e}}_1^T + \lambda_2 \hat{\boldsymbol{e}}_2 \hat{\boldsymbol{e}}_2^T + \cdots + \lambda_d \hat{\boldsymbol{e}}_d \hat{\boldsymbol{e}}_d^T$$

Covariance Matrix

$$(X-\mu)^T C^{-1}(X-\mu)$$

Mahalanobis Distance Expression

$$C = Q\hat{C}Q^T$$

Matrix Diagonalization

Interpretation

Mahalanobis Distance Revisited

$$(\boldsymbol{X}-\boldsymbol{\mu})^T \boldsymbol{C}^{-1}(\boldsymbol{X}-\boldsymbol{\mu}) = (\boldsymbol{X}-\boldsymbol{\mu})^T \left(Q\widehat{C}Q^T\right)^{-1}(\boldsymbol{X}-\boldsymbol{\mu})$$

$$= (\boldsymbol{X}-\boldsymbol{\mu})^T (Q^T)^{-1}\left(\hat{C}^{-1}\right)(Q)^{-1}(\boldsymbol{X}-\boldsymbol{\mu})$$

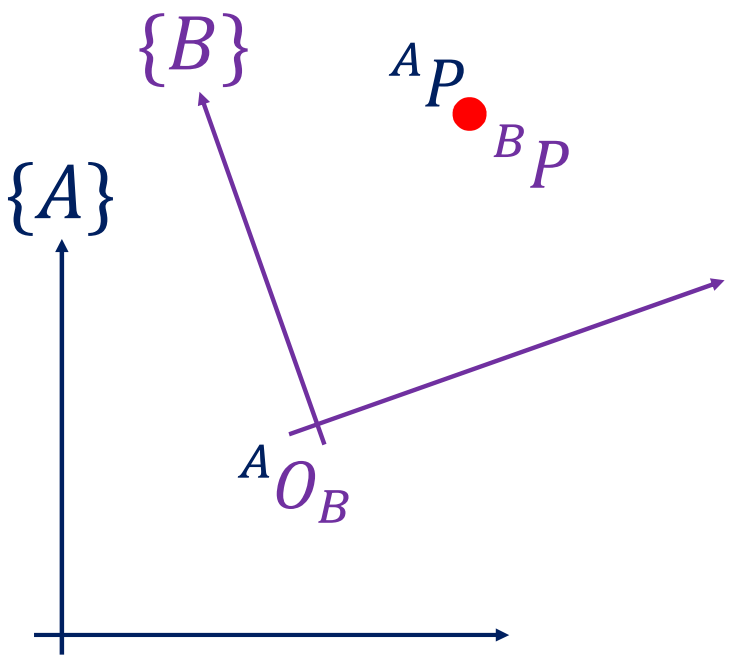
$$= (\boldsymbol{X}-\boldsymbol{\mu})^T Q\left(\hat{C}^{-1}\right)Q^T(\boldsymbol{X}-\boldsymbol{\mu})$$

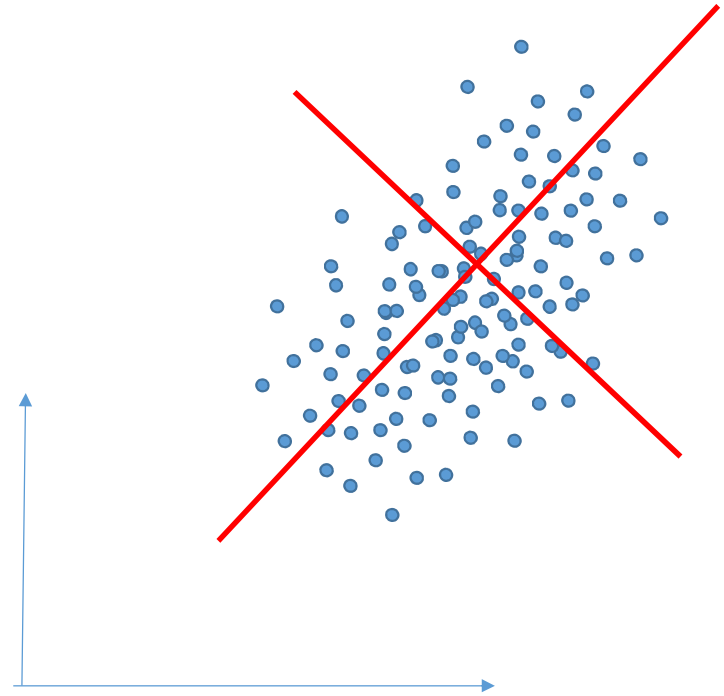
$$= \left(Q^T(\boldsymbol{X}-\boldsymbol{\mu})\right)^T\left(\hat{C}^{-1}\right)\left(Q^T(\boldsymbol{X}-\boldsymbol{\mu})\right) = \boldsymbol{U}^{\boldsymbol{T}}\hat{C}^{-\mathbf{1}}\boldsymbol{U}$$

$$= \sum_{i=1}^{n} \frac{(\boldsymbol{U}[\boldsymbol{i}])^2}{\boldsymbol{\lambda_i}}$$

$$\boxed{\boldsymbol{U} = Q^T(\boldsymbol{X}-\boldsymbol{\mu})}$$

Relating the Two





$${}^{B}P = {}^{A}_{B}R^{T}\left({}^{A}P - {}^{A}O_B\right)$$

$$\boldsymbol{U} = Q^{T}(\boldsymbol{X} - \boldsymbol{\mu})$$

Summary

- Coordinate Transformations
- Recalling Definitions
  - Covariance Matrix
  - Mahalanobis Distance
  - Matrix Diagonalization
- Covariance Matrix – An Interpretation

Thank You